



Taktwerk4RC

a DIY „*BeatBuddy*“ without internal Drum-Sounds

based on ESP32 S3 N16R8 DevKitC-1 Module

Status: Draft 25.9.2026

Key features of this „MIDI-File-Player“ ?

- USB-C-Host to drive the RC600 (not Power, only MIDI-Data), *no MIDI or Audio-Cables needed... Clean Connections*
- Receive Clock via USB-C, send MIDI-NoteOn/Off for Drums via USB-C
- Taktwerk4RC will loop all „loop“-files and play „fill“-files only once
- Seamless switching MIDI-Files on the "1": If you press the "Fill" footswitch in the middle of a bar, the song remains synchronized. The device waits until the start of the next bar (tick 0) to switch directly to the next `fill-n.mid` file.
- Dynamic time signatures: When the MIDI file is first opened, the meta-event parser reads the time signature specification (e.g., 3/4 or 6/8). The beat ticks and the NeoPixel metronome automatically adjust their counting accordingly.
- Real-time volume reduction: Using the optional `FOOT-QUIET` footswitch, the velocity byte of all NoteOn commands is halved within the `midiCallback`, allowing for an instantaneous switch to a quieter passage without latency.

SD-Card-Structure

Songs are Folders

/Song_01_Rock/

- loop-1.mid
- fill-1.mid
- loop-2.mid
- fill-2.mid

/Song_02_Waltz_3_4/

- loop-1.mid
- fill-1.mid

Song Next

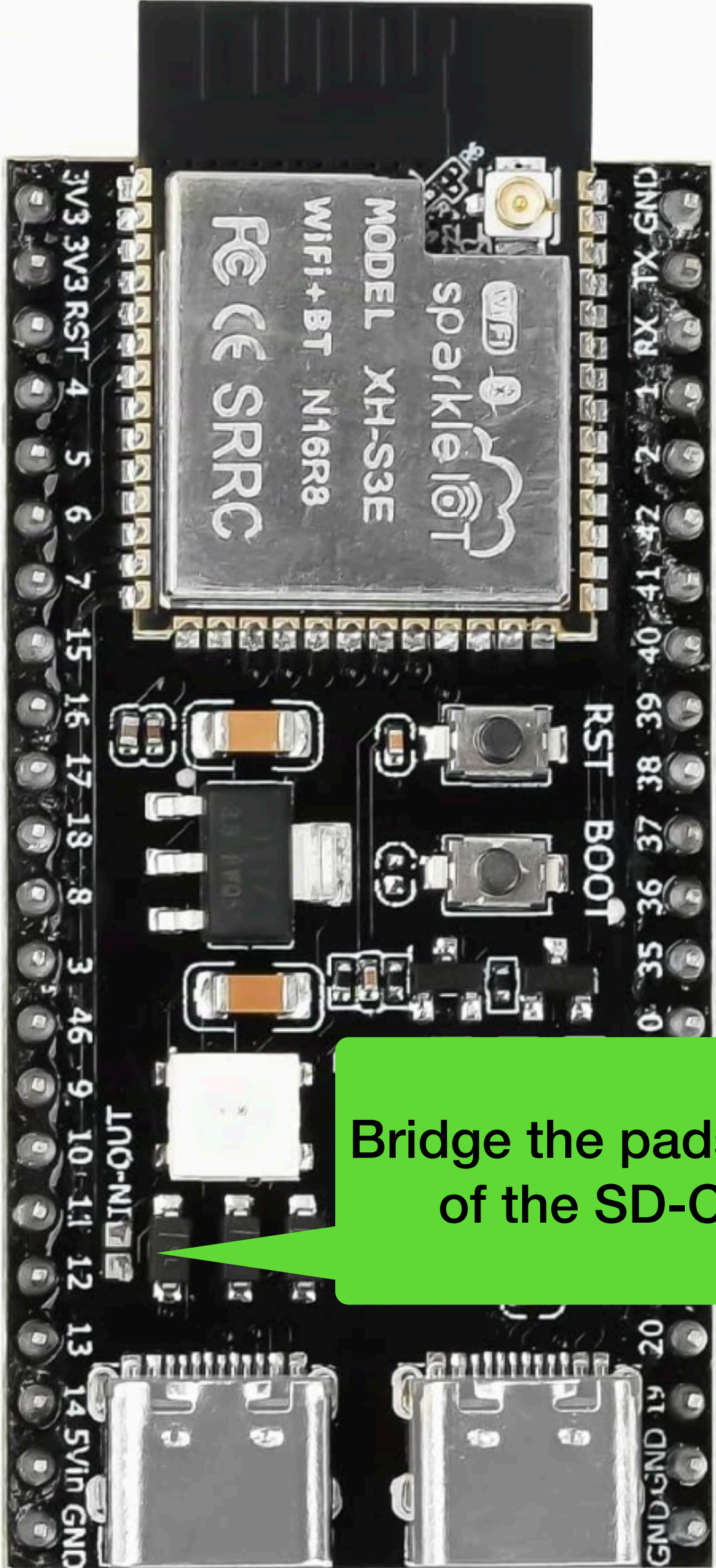
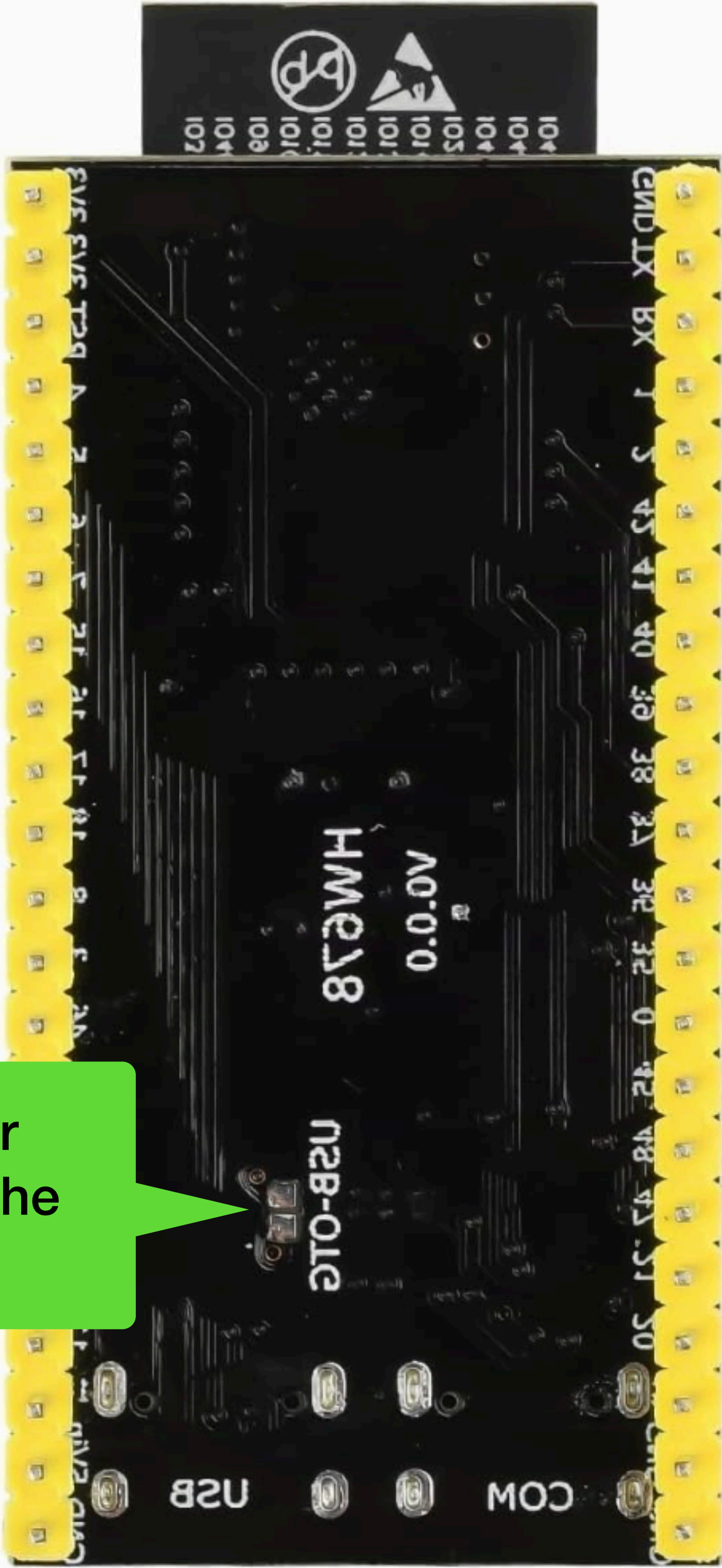
FS1



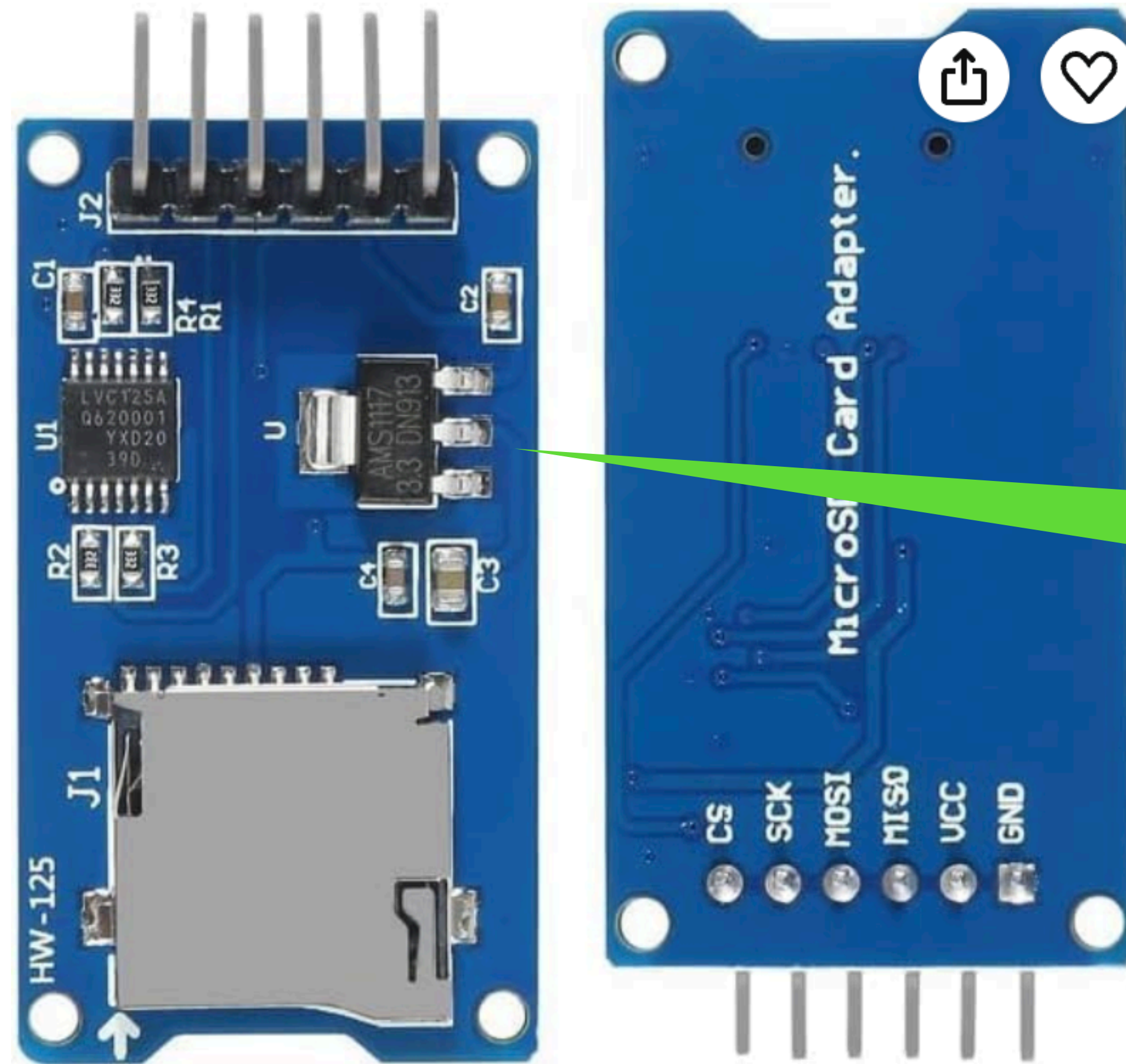
Pinout

Modul	Signal	ESP32-S3	GPIO	Bemerkung
I2C OLED (SSD1306)	SDA	GPIO 8		Standard I2C Data
	SCL	GPIO 9		Standard I2C Clock
SPI SD-Card	CS	GPIO 10		Chip Select
	MOSI	GPIO 11		SPI Master Out
	SCK	GPIO 12		SPI Clock
	MISO	GPIO 13		SPI Master In
Taster (Navi)	Song Next	GPIO 4		Interner Pullup (gegen GND)
	Song Prev	GPIO 5		Interner Pullup (gegen GND)
Fußtaster	Start / Stop	GPIO 6		Interner Pullup (gegen GND)
	Fill Trigger	GPIO 7		Interner Pullup (gegen GND)
	Quiet (50% Vol)	GPIO 15		Interner Pullup (Option/Schalter)
USB Host	D- / D+	GPIO 19	/ 20	Belegt für RC-600 USB
NeoPixel LED	DATA	GPIO 48		Onboard LED / Metronom

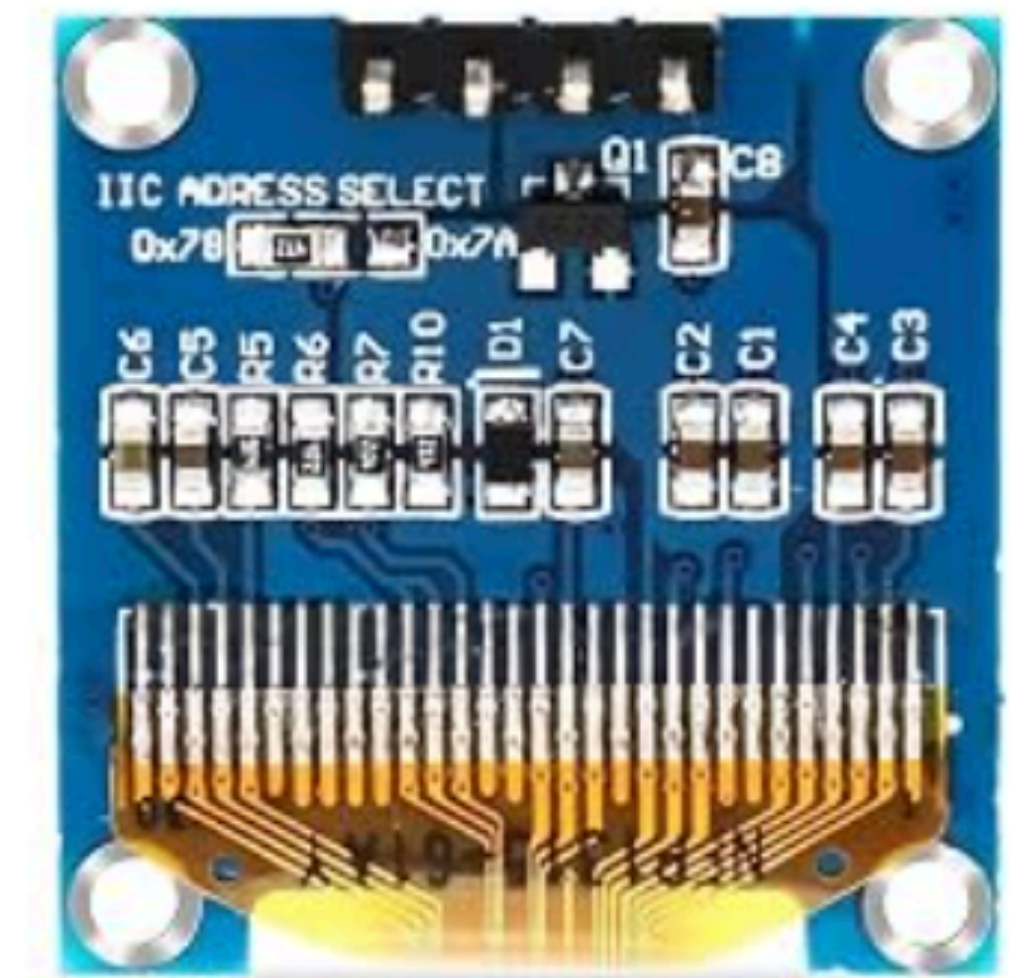
Pads connected



0,96 Zoll OLED Display Modul 128x64 Pixel I2C Display



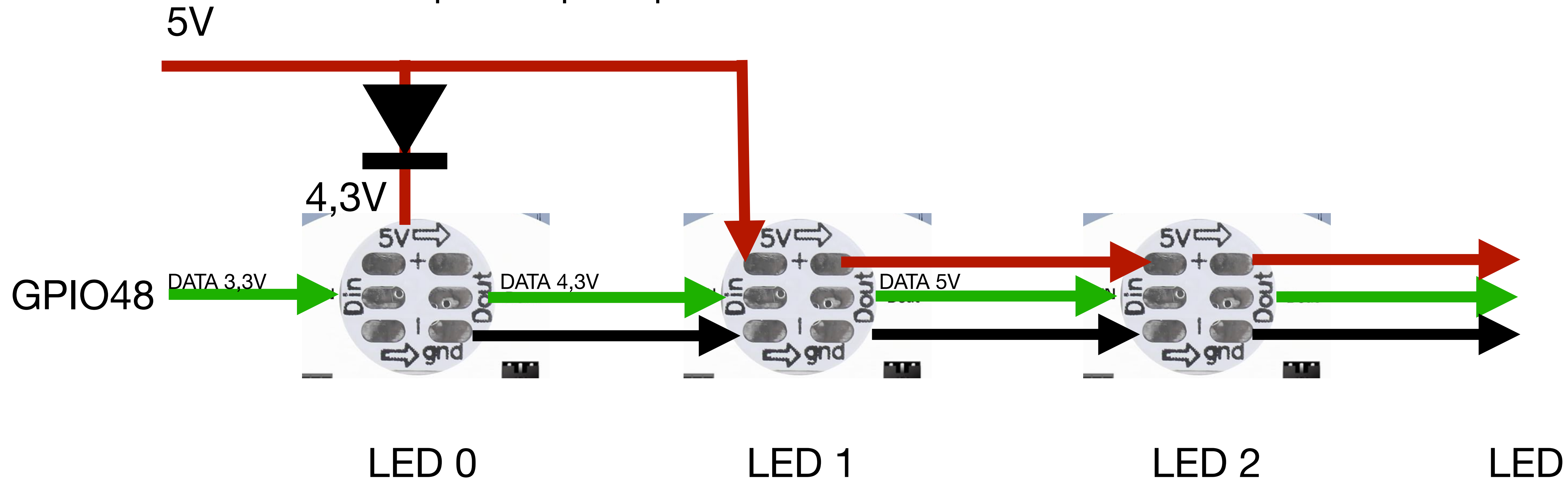
This SD-Card Module is 3.3 Volts compatible for the Data-Lines, but it needs 5 Volts on VCC because of the 3.3Volt Drop-Down-Regulator



3.3Volts Neopixel- Trick

If you want to control a 5V NeoPixel LED using a 3.3V data line, you can use a diode to lower the supply voltage for the first LED to 4.3 volts. This enables it to process 3.3V data signals.

The subsequent LED can be operated at 5 volts and will accept the data signals output by the first LED via its D-Out pin. ... perhaps someone would like to add more LEDs ...



Drum-Editor

Taktwerk-Editor

- German Editor
- Upload any MIDI-File
- Select the Channel with the Drums
- Select single or multiple Bars and export them as MIDI-Files with the Name „loop-*.mid“ or „fill-*.mid“
- use the Grid-Editor to edit a selected bar
- listen to single bars

Takte auswählen:

Takt 1
73 Noten
▶ Play

Takt 2
67 Noten
▶ Play

Drum Grid – Takt 1 Play Grid

Instrument	1	2	3	4	5	6	7	8	9	10
Bass Drum 1										
Tambourine										
Cowbell										
Hi Bongo										
Low Bongo										
Mute Hi Conga										
Open Hi Conga										
Low Conga										
Maracas										
Claves										

■ 0-29 ■ 30-49 ■ 50-74 ■ 75-99 ■ 100-109 ■ 110-127

☒ loop- ☐ fill- .mid Als MIDI Exportieren

Taktwerk - Standalone MIDI Inspector

Liest MIDI Formate 0 & 1 ohne externe Abhängigkeiten direkt aus.

MIDI-Datei hierhin ziehen oder **klicken** zum Auswählen

MIDI Format: 0 | Tracks: 1 | PPQN: 480

Spur 1

140 Noten | **Kanäle: 10**

Zu verarbeitender MIDI-Kanal:

Kanal 10 (Standard Drums) ▾

Tempo: 120 BPM MIDI-Tempo: 120.00 BPM

File-Selector for MIDI-Files

Analyzed Information based on selected MIDI-File and existing Channels

Best-Effort selection of a track 10 or if it doesn't exist, select the right MIDI-Channel.
(I had MIDI-Files with Drums in Channel 1)

Your tempo versus Tempo provided in MIDI-File

Select a single or multiple Bars
(Takt) for export
(Play does only 1 bar!)

Editor for selected bar

Export as loop or fill - .mid-File.
Taktwerk4RC will loop all loop-
files and play fill-file only once

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▶ Play

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Drum Grid – Takt 1

Play Grid

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0-29 30-49 50-74 75-99 100-109 110-127

☒ loop-

☐ fill-

1

.mid

Als MIDI Exportieren